ALONG A FLAT RING MAP INDUCING AN ISOMORPHISM MODULO A FINITELY GENERATED IDEAL, RESTRICTION IS AN EQUIVALENCE OF DERIVED COMPLETE DERIVED CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'Some details omitted.' at proof line 26662 and leaves the final sentence of the essential-surjectivity paragraph without a full stop at proof line 26669; both are reproduced unchanged. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Derived complete objects D_comp(A, I) and the vanishing of Ext out of the localizations A_f for f in I. the hypothesis at statement lines 26636-26637 and used at proof lines 26648-26650. Is what makes the Cech resolutions of A and B quasi-isomorphisms against the given K-injective representative.
- Extended alternating Cech complexes and the quasi-isomorphism comparing them for A and B. invoked at proof lines 26644-26646. Is the tool that transfers information from A to B along the flat map.
- K-injectivity, including its preservation under the restriction of scalars. used at proof lines 26664 and 26676-26677. Lets Hom_A(B, -) compute the correct derived objects and lets a B-module complex serve as an A-module complex.

Source reference. Original lemma 13926: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13926–13926. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 25850: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25850–25850. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 24363: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24363–24363. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $I \subset A$ be a finitely generated ideal. If $A \to B$ is flat and $A/I \cong B/IB$, then the restriction functor $D(B) \to D(A)$ induces an equivalence $D_{comp}(B, IB) \to D_{comp}(A, I)$.*

Proof. Choose generators $f_1, \ldots, f_r$ of I. Denote $\check{\mathcal{C}}^\bullet_A \to \check{\mathcal{C}}^\bullet_B$ the quasi-isomorphism of extended alternating Čech complexes of Lemma 24363. Let $K \in D_{comp}(A, I)$. Let $I^\bullet$ be a K-injective complex of A-modules representing K. Since $\mathrm{Ext}^n_A(A_f, K)$ and $\mathrm{Ext}^n_A(B_f, K)$ are zero for all $f \in I$ and $n \in \mathbf{Z}$ (Lemma 25850) we conclude that $\check{\mathcal{C}}^\bullet_A \to A$ and $\check{\mathcal{C}}^\bullet_B \to B$ induce quasi-isomorphisms

$$I^\bullet = \mathrm{Hom}_A(A, I^\bullet) \longrightarrow \mathrm{Tot}(\mathrm{Hom}_A(\check{\mathcal{C}}^\bullet_A, I^\bullet))$$

and

$$\mathrm{Hom}_A(B, I^\bullet) \longrightarrow \mathrm{Tot}(\mathrm{Hom}_A(\check{\mathcal{C}}^\bullet_B, I^\bullet))$$

Some details omitted. Since $\check{\mathcal{C}}^\bullet_A \to \check{\mathcal{C}}^\bullet_B$ is a quasi-isomorphism and $I^\bullet$ is K-injective we conclude that $\mathrm{Hom}_A(B, I^\bullet) \to I^\bullet$ is a quasi-isomorphism. As the complex $\mathrm{Hom}_A(B, I^\bullet)$ is a complex of B-modules we conclude that K is in the image of the restriction map, i.e., the functor is essentially surjective

In fact, the argument shows that $F : D_{comp}(A, I) \to D_{comp}(B, IB)$, $K \mapsto \mathrm{Hom}_A(B, I^\bullet)$ is a left inverse to restriction. Finally, suppose that $L \in D_{comp}(B, IB)$. Represent L by a K-injective complex $J^\bullet$ of B-modules. Then $J^\bullet$ is also K-injective as a complex of A-modules (Lemma 13926) hence $F(\text{restriction of } L) = \mathrm{Hom}_A(B, J^\bullet)$. There is a map $J^\bullet \to \mathrm{Hom}_A(B, J^\bullet)$ of complexes of B-modules, whose composition with $\mathrm{Hom}_A(B, J^\bullet) \to J^\bullet$ is the identity. We conclude that F is also a right inverse to restriction and the proof is finished. □